

Spectra Talk

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1	The motivational section	1
2	What is a spectrum	3
2.1	Rigidity	4
3	Properties of Sequential Spectra	5
3.1	Relating them to spaces	5
3.2	Cohomology Theories	7
4	Another motivational section	8
5	A sketch of the ‘modern perspective’	8

The moto for spectra is making topology algebraic.

- loop spaces are the adjoints of spectra, are coherent monoids in top
- Stable homotopy is a homology while normal homotopy is not
- The stable category is triangulated
- They represent homology theories

We reference [DK01], [Swi75], [DS95] and [BR20]. We will just assume that Top refers to a convenient category for topological spaces (we want the smash to have an adjoint).

1 The motivational section

This is mostly vibe history. I guess the earliest example of stable phenomena is Freudenthal's suspension theorem:

Lemma (A corollary, 1937). *For any X the sequence*

$$\pi_i(X) \rightarrow \pi_{i+1}(\Sigma X) \rightarrow \pi_{i+2}(\Sigma^2 X) \rightarrow \cdots$$

eventually stabilises, that is the maps all become isomorphisms.

Proof. Suspending a space increases its connectivity by one and Freudenthal actually showed that for an n connected space the maps are iso for $i < 2n$ (the proof of this is decomposing the suspension into cones and observing that the inclusions are equivalences and applying adjunction to see that the degree has shifted). Thus the range in which it's an iso doubles each time you suspend.

This leads to the definition of the stable homotopy groups of a space as

$$\pi_i^s(X) := \operatorname{colim}_k \pi_{k+i}(\Sigma^k X).$$

For $X = S^0$ these are commonly called the stable stems (for other spheres they are just zero). Because suspending is just smashing with spheres and suspending a sphere gives you one higher sphere we can see that the stable homotopy groups are exactly $\pi_i^s(X) = \operatorname{colim}_k [S^{i+k}, S^k \wedge X]$.

Soon after this Pontryagin began his work on what is now known as the Pontryagin-Thom construction. Apparently this work was over the course of several decades, but the core idea was perhaps established early

Lemma (1937-ish). *There is a bijection*

$$\Omega_{k-n,M}^{\operatorname{fr}} \leftrightarrow [M, S^n]$$

where $\Omega_{k-n,M}^{\operatorname{fr}}$ is the space of $k - n$ dimensional framed submanifolds of M up to framed co-bordism. Moreover

$$\Omega_k^{\operatorname{st}+fr}(X) \cong \pi_k^s(X_+)$$

where the group on the left is that of stably framed bordism classes of k dimensional manifolds over X . Other bordism theories, for any stable structure on a vector bundle, are also representable in this way.

Proof. The bijection in the forward direction is given by taking a submanifold with a framing (N, ν) to the so called collapse map, $M \rightarrow S^n = \mathbb{R}^n \cup \infty$ given by sending everything outside of N to ∞ and everything in N to its framing (which is a vector in $\mathbb{R}^n$). In the reverse it is given by taking a map $f : M \rightarrow S^n$ and simply looking at $f^{-1}(r)$ for r some regular value, this submanifold has a framing by pulling back the standard framing on the sphere.

In the 1940's - 1950's Eilenberg and MacLane made several relevant observations to our story, in a modern form we can state them as

Lemma (1954-ish). *Now we denote $K(\mathbb{Z}, i)$ the Eilenberg-MacLane space, only non-trivial homotopy group is $\mathbb{Z}$ in degree i , then*

$$H^n(X; \mathbb{Z}) \cong [X, K(\mathbb{Z}, n)]$$

and

$$H_n(X; \mathbb{Z}) = \operatorname{colim}_k [S^{n+k}, X_+ \wedge K(\mathbb{Z}, k)]$$

where $X_+ = X \sqcup *$.

Notice that there is an asymmetry between cohomology and homology. This is somehow a manifestation of the fact that covariant things in Spectra are (stable) homotopy groups while contravariant things are cohomology.

Shortly after this Thom made his contribution, removing the framing condition he constructed spaces $MO(n)$ such that

Lemma (1954).

$$\Omega_k \cong \operatorname{colim}_n \pi_{n+k} MO(n)$$

Which moreover assembles into an isomorphism of rings.

The sphere from this perspective is the Thom space of the trivial bundle over a point, more generally $MO(n)$ is the Thom space of the universal bundle $EO \rightarrow BO$. He further classified the cohomology of $MO(n)$ using Stiefel-Whitney classes, making this guy more or less computed.

Using Eilenberg and Steenrods axiomatisation of homology, and inspired by Thom's MO (G. W.) Whitehead made the first attempt at introducing 'spectra' (the now defunct Spanier-Whitehead category) and proved

Lemma (1962). π_i^s is a (extraordinary) homology theory.

Recall that extraordinary means that it does not satisfy the dimension axiom. Note the difference between bordism and stable homotopy groups, they are both homology theories but stable homotopy is *reduced*.

Remark. (Sequential colimits in the category of groups) A colimit is something like a pushout, or a disjoint union. A sequential colimit collects all the groups in the sequence (takes the disjoint union) and then quotients by the relation coming from the maps, namely two elements are declared the same if their image ‘somewhere’ or ‘eventually’ in the sequence is identified.

Remark. (Summary) What we observe from Freudenthal is that we can always create a sequence of spaces from a single space by taking the suspension of the space iteratively, and that moreover there is a sensible notion of homotopy group for such a thing. What Pontryagin-Thom and Eilenberg-MacLane show is that there are natural examples of such sequences, namely $S^n, K(\mathbb{Z}, n)$ and $MO(n)$ that moreover seem to represent geometric or homological information. Finally Whitehead attempts to axiomatise these things as Spectra and makes a precise relation between homology theories and representability in terms of spectra.

2 What is a spectrum

The notion of a spectrum is one developed by homotopy theorists. This means that what a spectrum *really* is, is a isomorphism type in a *homotopy* category. What we give here is a point set construction over the homotopy category. The standard thing is to then put a model structure on this point set model and consider the localisation of the category. The point of this manouver from my perspective is two fold. One is that it is definitionally easier to state the point set models, the other is mainly computational. Because when you localise the objects stay the same, we are at liberty to call the objects of both the unlocalised and localised category ‘spectra’ as long as we bear in mind that what people really care about is only spectra up to homotopy equivalence (the analogue of homeomorphism for spectra is not interesting to people as far as I can tell).

We will now define several categories of spectra, and claim that they have model structures such that their homotopy categories, known as the *stable category*, are all equivalent.

Spectrum. Let $G_n = *, O(n), \Sigma_n$. The category of G -spectra, denoted $\mathcal{S}^G$, has objects $E = \{G_n \curvearrowright E_n, \sigma_n\}_{n \in \mathbb{N}}$ sequences of spaces with continuous G_n action paired with maps $\sigma_n : \Sigma_n E_n \rightarrow E_{n+1}$. The suspension and action are further required to be compatible in the sense that the map, for any n, k

$$\Sigma^k E_n \xrightarrow{\sigma^k} E_{n+k}$$

has to be $G_{n \times k}$ equivariant, where $G_n \times G_k$ is considered as a subgroup of G_{n+k} in the natural way, which acts on the left by suspending the action on E_n and on the right by definition.

A morphism $f : E \rightarrow F$ is a collection of G_n equivariant maps $f_n : E_n \rightarrow F_n$ such that the obvious diagram commutes

$$\begin{array}{ccc} S^1 \wedge E_n = \Sigma E_n & \xrightarrow{\text{id} \wedge f_n} & \Sigma F_n \\ \downarrow \sigma & & \downarrow \sigma \\ E_{n+1} & \xrightarrow{f_{n+1}} & F_{n+1} \end{array}$$

Sequential Spectra. Lets focus on the $G = *$ case. In this case the category we have defined is called ‘sequential spectra’ and also more typically denoted $\mathcal{S}^{\mathbb{N}}$. The equivariance conditions mean nothing and so a sequential spectrum is just a collection of spaces with comparison maps between the $n + 1$ -th term and the suspension of the previous term. Note that there are obvious forgetful functors from $\mathcal{S}^{\Sigma}, \mathcal{S}^O \rightarrow \mathcal{S}^{\mathbb{N}}$. These other two categories are called sequential and orthogonal spectra respectively.

We define the homotopy groups of a sequential spectrum as a colimit

$$\pi_i(E) := \operatorname{colim}_k \pi_{i+k}(E_k)$$

Then the notion of weak equivalence between spectra is the same as Top, a map of spectra that induces an isomorphism on all homotopy groups. We then define the so called stable homotopy category as the localisation of sequential spectra at these weak equivalences [BR20, Thm 2.3.12]. **Thats it, thats what a spectrum is.**

Example. The sphere spectrum is $\mathbb{S} = \{S^n\}_n$ where the structure maps are the homeomorphisms $\Sigma S^n \rightarrow S^{n+1}$.

Example. $\mathbb{K}(\mathbb{Z}) = \{K(\mathbb{Z}, n)\}_n$ whose structure maps are given by the adjoint of the homotopy equivalence $\Omega K(\mathbb{Z}, n+1) \simeq K(\mathbb{Z}, n)$, to be specific

$$h \in [K(\mathbb{Z}, n), \Omega K(\mathbb{Z}, n+1)] \cong [\Sigma K(\mathbb{Z}, n), K(\mathbb{Z}, n+1)]$$

by the adjunction.

A spectra is called an **Ω -spectrum** if the adjoint maps to the structure maps are all (weak) homotopy equivalences, these are the *fibrant objects*. It is called **CW-spectrum** if every space in the spectrum is a CW complex and the structure maps are inclusions of subcomplexes, these are the *cofibrant objects* in the unstated model structure. A spectra with only finitely many negative homotopy groups is called **bounded below**, with no negative homotopy groups is called **connective-spectrum** and a connective spectrum with $\pi_0 = 0$ is called a **connected-spectrum**. Every loop space defines a connected Ω spectrum by applying Σ^∞ defined below, in particular we deloop it at each step and the structure map is just a chosen homotopy equivalence (which exists because it is an infinite loops space).

It should be noted that by general model category theory everything in the homotopy category is isomorphic to a bifibrant object, in particular for spectra, up to homotopy every spectrum is equivalent to an Ω -CW spectrum.

Remark. This category has an explicit description of the morphisms given by so called “cofinal” maps. This is a little clunky so we will just sketch it. For two CW spectrum, spectrum where every space is CW complex and the structure maps are inclusions of subcomplexes (they are the fibrant objects) a subspectrum $F \subseteq E$ is called cofinal if each cell of the spectrum (a cell in some space), under the inclusion into the higher levels of the spectrum, eventually ends up in F . Morally E is sort of becoming F in the limit. We can now describe the morphisms $E \rightarrow F$, let

$$S = \{(E', f') : E' \subseteq E \text{ cofinal}, f' : E' \rightarrow F \text{ a function}\}$$

be maps from cofinal subcomplexes into F . Then we say that two such maps are equivalent $(E', f') \sim (E'', f'')$ iff there is an (E''', f''') cofinal in $E' \cap E''$ such that f', f'' and f''' agree on E''' . (something like the germ of the functions). Composition can just be defined by picking representatives that make sense to compose.

2.1 Rigidity

Now the model structures on the other categories of spectra can be more complicated but there are constructions. In [BR20, 1.1.4] gives the following list of desiderata for a stable homotopy category $\mathcal{SHC}$, that is a category that captures all of our stable phenomena. The properties are as follows

1. There should be a natural adjunction

$$\Sigma^\infty : \operatorname{HoTop} \rightleftharpoons \mathcal{SHC} : \Omega^\infty$$

2. Given two spaces A, B there should be a bijection

$$[\Sigma^\infty A, \Sigma^\infty B] \cong [A, B]^{stab} := \operatorname{colim}_k [\Sigma^k A, \Sigma^k B]$$

between homotopy classes of maps of their spectra and *stable* homotopy classes of maps of the spaces.

3. For every (co)homology theory there is a unique up to isomorphism object that represents theory.
4. A map in $\mathcal{SHC}$ should induce a natural transformation on the (co)homology theory.
5. Various nice categorical properties, such as being monoidal, cartesian closed and being enriched over graded abelian groups.

These are obviously all very reasonable and desirable, moreover encapsulate stability. Either explicitly requiring hom sets to be stable hom sets for spaces, or by requiring it to be representing cohomologies, which by our examples are somehow given by stable homotopy groups of smashing with spaces. Now the theorem is that the category of sequential spectrum localised at weak equivalences satisfies all of these properties. There is also a type of uniqueness, called rigidity

Theorem ([BR20], 5.7.1). *If $\mathcal{C}$ is a stable model category and there is an equivalence of triangulated categories*

$$\mathcal{SHC} \rightarrow \operatorname{Ho}\mathcal{C}$$

then $\mathcal{C}$ is Quillen equivalent to sequential spectra.

We wont define any of this here but the point is that if two model categories present the stable homotopy category as its homotopy category then they are equivalent. Note that it doesnt say that if you satisfy *these* axioms you are equivalent to the stable homotopy category as defined above, merely that the stable homotopy category is one category satisfying these things and any model category representing *this* category are equivalent.

Remark. The point of the other point set models is that they make the properties on the homotopy category more or less obvious. Sequential spectra famously have a very hard time being monoidal, while symmetric and orthogonal have a monoidal structure that descends.

3 Properties of Sequential Spectra

So far what have we done. We have seen that historically spectra arose out of stable phenomena and it was observed that these stable phenomena were heavily connected to the algebraic side of topology through (co)homology theories. A point set model for a category of spectra was then given and further equipped with a notion of weak equivalence. We claimed that the associated homotopy category made precise the connection to both stable phenomena, through classes of maps, as well as cohomology theories, through representability. Moreover we claimed that the point set models are essentially unique, although technically usefull to have options. Lets now see some of the details of these claims as they relate to cohomology, the stability claims are clear by the definition of weak equivalences.

3.1 Relating them to spaces

First there are many natural functors that should simply be observed. The most important is the (Quillen) adjunction called the (shifted) suspension / evaluation, which mirrors the adjunction in Top given by $[\Sigma X, Y] \cong [X, \Omega Y]$,

$$F_d : \operatorname{Top} \dashv \mathcal{S}^{\mathbb{N}} : \operatorname{Ev}_d$$

$$X \mapsto \begin{cases} \Sigma^{n-d} X, & n \geq d \\ *, & \text{else} \end{cases}$$

$$K_d \hookleftarrow \{K_n\}_n$$

The left adjoint is faithful, the derived functors, those induced on the homotopy category are not faithful, as some maps may be identified after repeated suspension. We denote $\Sigma^\infty = F_0$, this can also be thought of as the left derived functor of the above. The right derived functor is denoted Ω^∞ . Moreover it is clear that

$$\pi_n(F_0(X)) = \pi_n^s(X)$$

the stable homotopy groups.

Lemma. $\Omega^\infty E$ is an infinite loop space.

Proof. It is the zeroth space of a spectrum E and because we considered it as the derived functor we assume that the spectrum is an Ω -spectrum. Hence the first structure map $\Sigma E_0 \rightarrow E_1$ has an adjoint $E_0 \rightarrow \Omega E_1$ which is an equivalence, and on we have $E_0 \rightarrow \Omega E_1 \rightarrow \Omega \Omega E_2$ etc.

One thing to note is that there is not an equivalence

$$\Omega^\infty \Sigma^\infty X \not\simeq \text{colim}_n \Omega^n \Sigma^n X$$

strictly. On the other hand when the functors are *derived* this equivalence holds!

Remark. (Loop spaces are equivalent to spectra) It is often said that infinite loop spaces are equivalent to Ω -spectra. One needs to specify however that an infinite loop space is a space with *specified* deloopings and that moreover a map of such infinite loop spaces is a map that commutes with all the specified deloopings in a coherent way. From this setup it's therefore almost a tautology. In particular it is *not* the case that for a topological space that happens to be an infinite loop space the relation to spectra is at all concrete, neither of the functors above are faithful on Top .

Warning. Unrelated (well later we will see some relation) to these suspension and loop functors is the functor given by wedging and homing to a topological space. Let A be an ordinary space and E be a spectrum, then

$$E \wedge A := \{E_n \wedge A\}_n$$

$$\text{Hom}(A, E) := \{\text{Hom}(A, E_n)\}_n$$

which gives the loop and suspension of spectra $\Sigma E = E \wedge S^1$ and $\Omega E = \text{Hom}(S^1, E)$. These two form a Quillen adjunction

$$\Sigma : \mathcal{S}^\mathbb{N} \rightleftarrows \mathcal{S}^\mathbb{N} : \Omega$$

that is moreover an equivalence. Notice that this is an equivalence from spectra to spectra, whilst the previous is an adjunction between spectra and spaces.

Remark. We might have indexed by $\mathbb{Z}$, however there is again a Quillen equivalence

$$\mathcal{S}^\mathbb{N} \rightleftarrows \mathcal{S}^\mathbb{Z}$$

given by truncating in the backward direction and extending into the negative degrees by the point in the forward direction.

Remark. Lewis ‘Is there a convenient category of spectra?’ proves that under some mild hypotheses no strict category of spectra exists that satisfies five axioms, namely has a smash, a $\Omega^\infty, \Sigma^\infty$ adjunction, unit of the smash is the sphere spectrum, the weak equivalence $\Omega^\infty \Sigma^\infty X \simeq \text{colim}_n \Omega^n \Sigma^n$ and $\Omega^\infty, \Sigma^\infty$ distribute over the smash. The problem is making all the operations strict. We can still have all of this in *the stable category* (deriving the spaces and functors).

3.2 Cohomology Theories

Lemma (Brown Representability, 1962). *A functor $F : \text{hoCW}^{\text{op}} \rightarrow \text{Set}$ that maps coproducts to products and maps homotopy pushouts to ‘weak’ pullbacks (writing this out explicitly is saying that from a covering by two subcomplexs (pushout) we can compute the functor given the overlap (pullback)); Mayer-Vietoris) is representable by a CW complex C , that is $F \cong [-, C]$.*

The point being that a representable functor was already known to satisfy the conditions and therefore this gives a necessary and sufficient condition for representability. These conditions are essentially Mayer-Vietoris and additivity (as well as homotopy invariance) and so are satisfied in particular by any (extraordinary) (co)homology theory. Thus *level wise* each cohomology is representable by a CW complex.

Now given a spectrum $\{E_n\}$ then we have a reduced cohomology theory given by

$$h^n(X) = \text{colim}_k [X \wedge S^k, E_{n+k}] = \text{colim}_k [X, E_n] = [X, E_n]$$

applying adjunction and assuming that E is an omega spectrum (because as we have said this is the case up to weak equivalence). Alternatively this can be written as

$$h^n(X) = [\Sigma^\infty X, \Sigma^n E] = \pi_{-n}[\Sigma^\infty X, E]$$

using that Σ is an auto equivalence of the category of spectra. Note that here we are taking π_{-n} of the internal hom spectra, defined by the adjoint to the monoid structure on Sp . These formulas therefore imply it would make sense to consider the cohomology of a *spectrum* as well as a space. A reduced homology theory is given by

$$h_n(X) = \text{colim}_k [S^{n+k}, X \wedge E_k] = \pi_n(X \wedge E)$$

This makes the duality between cohomology and homology most explicit in spectra. We can make the theories unreduced by just precomposing with $X \mapsto X_+$. Note this is not the construction of a co/homology *theory*. What we have here are just the *groups*, to get the theory, one needs certain LES’s to exist. This is where the structure maps of the spectrum are used, not just the spaces. We don’t do it here.

The final step is to see that all cohomology theories are represented by Spectra.

Lemma ([BR20], Cor 1.1.26). *All cohomologies are represented.*

Proof. Let h^n be a cohomology theory. Take the sequence of spaces given to you level wise by Brown representability, which are unique up to homotopy and can be chosen to be connected, K_n . Then by suspension isomorphism on cohomology and adjunction we have that

$$h^n(K_n) = [K_n, K_n] \cong h^{n+1}(\Sigma K_n) = [\Sigma K_n, K_{n+1}] \cong [K_n, \Omega K_{n+1}]$$

the identity in $[K_n, K_n]$ maps to a comparison map in $[K_n, \Omega K_{n+1}]$, and can be seen to be a weak equivalence using a similar trick but inputting S^n instead of K^n above. This therefore defines a spectrum, which clearly represents this cohomology.

Example. We have already seen that $\Omega_k^{fr+st}(-)$ is the unreduced theory associated to the sphere spectrum, while $\pi_*^s(-)$ is the reduced theory. There is also $\mathbb{K}(\mathbb{Z})$ which represents ordinary cohomology and homology. Finally topological K theory is represented by $\mathbb{Z} \times BU$ in even degrees and $\Omega(\mathbb{Z} \times BU)$ in odd degrees.

Remark. (Point set failure) At the level of sequential spectra what happens. The point is that there are ‘not enough isomorphisms’. Concretely this means that not enough sequential spectra are identified, many sequential spectra will represent the same cohomology theory. So for representing cohomology sequential spectra is not ‘right’ either. From the point of view of identifying spaces up to stable weak equivalence the fact that we have to go to the stable homotopy is a tautology. It is just an interesting observation that the same relation is able to capture cohomology also.

4 Another motivational section

The first suitable from a modern perspective category of spectra was given by Boardman in 1963 and popularised by Adams in 1974 who gave the category a monoidal structure. Parallel to this development was the new age of algebraic geometry initiated by Alexander Grothendieck around 1958. One of Grothendieck's students Verdier wrote his thesis on the concept of a 'triangulated categorie', parallel to the publishing of the EGA books, in again around 1963. Verdier was inspired by earlier work in algebraic topology, with cofibrations being the model for an exact triangle. A triangulated category is an additive category with a 'shift' functor Σ (an autoequivalence) as well as some family of exact triangles $X \rightarrow Y \rightarrow Z \rightarrow \Sigma X$ that satisfy a list of axioms, it supposed to be a weakened version of abelianness. The point is that in algebraic geometry you study the rings or schemes via their derived categories, which are basically the homotopy category of the category of chain complexes. These derived categories have a triangulated structure that allows much of the formal algebraic arguments to go through. A triangulated category has an associated t-structure, basically a left and right half of the category, loosely it is a pair of full subcategories of D , $(D^{\leq}, D^{\geq})$, satisfying some properties. The heart of a t-structure is the intersection of these two subcategories and it is a fact that if you started with a category of modules then the heart of the derived category of that category of modules is again equivalent to the category of modules (algebra geometrically these are bundles). This is why derived categories are useful, they preserve at least the category of bundles over a scheme. In the 1970's it was realised that the category of spectra was triangulated in this sense. This was another connection drawn between the homotopy category in stable algebraic topology and algebra (ic geometry).

Next the work of May showed that infinite loop spaces are the topological analogue of commutative monoids.

Lemma (Recognition principle, 1972). *A (connected) space admits the structure of an infinite loop space (and hence is an Ω spectrum) iff it is an algebra over the $\mathbb{E}_{\infty}$ operad.*

Being an algebra over $\mathbb{E}_{\infty}$ is a precise statement of 'begin a coherent commutative monoid'. Thus here we have the principle that being a homotopical group is the same as being a spectrum.

In the late 90's the category of symmetric spectra allowed the construction of strictly commutative models of the smash product for spectra. This improves on Adams who is only commutative up to coherent homotopy. Finally in the 2000's Jacob Lurie writes higher algebra, giving a foundation on which infinity categories could be coherently discussed and showing many of the fundamental properties about the infinity category of spectra that we will try to sketch.

5 A sketch of the 'modern perspective'

The modern perspective as I have had it explained to me is more or less convincing depending on your tastes. There are several different constructions definition etc. The story is as complicated as the one above. Here is some stuff from Lior and Lurie's Higher Algebra.

Lior's Perspective. This is the story I was told by Lior, he insists that one should think of spectra as the homotopy version of commutative groups. Math starts out in sets, that is with no structure. There are two directions that it generalises, the topological (homotopical) and the algebraic. These generalisations can be fit into a conceptual and also functorial diagram **why do we take π_0 and not forget the topology?**

$$\begin{array}{ccc}
 \text{Set} & \xrightarrow{\text{discrete}} & \mathcal{S} \\
 \uparrow \text{forget} & \begin{array}{c} \perp \\ \leftarrow \pi_0 \end{array} & \uparrow \\
 \vdash & \mathbb{Z}[-] & \\
 \downarrow & & \downarrow \\
 \text{Ab} & \xleftarrow{\hspace{1cm}} & ?
 \end{array}$$

Spectra should be the thing that does *both*. An obvious candidate would be the infinity category $\mathcal{D}(\text{Ab})$, the derived category of abelian groups. The derived category is an example of so called ‘animation’,

$$\mathcal{D}_{\geq 0}(\text{Ab}) \simeq \text{Ani}(\text{Ab}) = \text{Fun}^{\times}(\text{Ab}_{\text{free, f.g.}}^{\text{op}}, \mathcal{S}).$$

This is (basically) just the free (sifted) cocompletion of $\text{Ab}_{\text{free, f.g.}}^{\text{op}}$, and comes from the one categorical motivation that for instance $\text{Ab} \simeq \text{Fun}^{\times}(\text{Ab}_{\text{free, f.g.}}^{\text{op}}, \text{Set})$. This is not the most homotopical thing that we could have done however, as we have just replaced the codomain of our functors with spaces, what about the domain?

To avoid technical difficulty we will talk now about the category CMon of commutative monoids. This also has a free finitely generated subcategory that can be conflated with $\mathbb{N}$, the number of generators. The hom sets can be seen to be matrices $\text{Hom}(n, m) = \text{Mat}_{n \times m}(\mathbb{N})$. Whenever a homotopy theorist sees $\mathbb{N}$ their eyes light up and they say “ Fin^{iso} (category of finite sets with only isomorphisms as morphisms) is the free commutative monoid in spaces” and $\pi_0 \text{Fin}^{\text{iso}} \cong \mathbb{N}$. Thus $\mathbb{N}$ is a decategorifies Fin^{iso} . Now we know what we need to do, we need to replace $\mathbb{N}$ by Fin^{iso} . We make the hom sets then $\text{Mat}_{n \times m}(\text{Fin}^{\text{iso}})$, where now we add and multiply finite sets using disjoint union and cartesian product. These additions and products will only associate up to isomorphism and therefore the category $\text{Mat}(\text{Fin}^{\text{iso}})$, with objects $\mathbb{N}$ and morphisms as described, is naturally a $(2, 1)$ -category. Thus we can generalise the animation process to define for instance

$$\text{CMon}(\mathcal{S}) := \text{Fun}^{\times}(\text{Mat}(\text{Fin}^{\text{iso}}), \mathcal{S})$$

where the functors obviously are as infinity categories. To get back to abelian groups we just take the group like objects in this category, and this gives us

$$\text{CGrp}(\mathcal{S}) := \text{CMon}(\mathcal{S})^{\text{gp}}$$

which is the model of Sp^{cn} connective spectra. This gives the naive idea “spectra are commutative homotopical groups”.

This is the sort of algebraic to homotopical angle to define spectra, from the stabilisation perspective, that is stabilising the category of spaces, we know however that this category of connective spectra sits inside a much larger category. In particular the sphere spectrum is not connective. From the topological perspective, we want to start with a space and go down in our diagram which is sort of increase the amount of algebraic structure it has, that is we want to see the homotopical to algebraic perspective. Topologically algebraic objects come from *loop spaces*. In finite loop spaces are the most commutative objects, with their homotopy commutative homotopy coherent operations satisfying all higher commutativity relations. Classically an infinity loop space can be seen to be equivalent to a connective sequential spectrum. This definition generalises straight away to the infinity categorical perspective as we can set

$$\text{Sp} := \lim_{\leftarrow} (\cdots \xrightarrow{\Omega} \mathcal{S} \xrightarrow{\Omega} \mathcal{S} \xrightarrow{\Omega} \mathcal{S})$$

This is the full category of spectra, as we have that

$$\text{Sp}^{\text{cn}} := \lim_{\leftarrow} (\cdots \xrightarrow{\Omega} \mathcal{S}^{1-\text{cn}} \xrightarrow{\Omega} \mathcal{S}^{0-\text{cn}} \xrightarrow{\Omega} \mathcal{S}).$$

Remark. (Terminology) It seems that what infinity category people call a spectrum is what [?] might call an Ω -spectrum. Recall that Σ^{∞} sends the category of infinite loop spaces to connected spectra. The limit we have defined unpacked defines an Ω -spectrum, which can be neither connected nor connective in general.

Higher algebra. So from this perspective we can see that the category of Spectra in the higher world is a natural setting for algebra, moreover we can see that the derived category of a ring lives in there. Well one of the points of higher algebra is that this is in some sense faithful. Lurie shows the following

Lemma (Prop 7.1.3.18[?]). *Given a connective $\mathbb{E}_{k+1}$ -ring R then there is an equivalence*

$$\mathrm{Alg}_R^{(k)} \simeq N\mathrm{Alg}_{\pi_0 R}$$

A similar theorem applies to the derived category, by page 2 of these notes, we have that for a discrete A_∞ ring spectrum R there is an equivalence of infinity categories

$$\mathrm{LMod}_R^{\mathrm{perf}} \simeq \mathcal{D}^{\mathrm{perf}}(\pi_0(R))$$

Thus if we have a ring spectrum that is discrete we can recover the derived category of the ordinary ring or the category of modules over that ring (that is the abelian category) that is its π_0 . On the other hand we should have a way to *produce* a discrete ring spectrum from an ordinary ring, and these should be inverses.

Definition (Ex 1.3.4, Ex 5.3.2, [?]). *Let R be an ordinary ring (commutative and unital) then HR is the sequential spectrum, in which the spaces are just the Eilenberg-MacLane spaces for the ring R .*

By [?, Ex 2.2.4] the Eilenberg-MacLane spectrum is discrete with only non-zero homotopy group in degree zero, given by the ring. The Eilenberg-MacLane spectrum is a ring spectrum by [?, Ex 5.3.2].

Thus for an abelian category, or a ring, we can either take the derived category or we can create an Eilenberg-MacLane spectrum from it. We have seen that the derived category construction allows us to recover the ring. Now we have seen that taking the EM spectrum and its category of modules (as spectra) allows us to recover *the derived category* and hence recover the original ring or abelian category. Thus ring spectra and modules over ring spectra are a suitable generalisation of rings and modules!

Remark. Using this it is possible to see how a ring is generalised by a ring spectrum. Since $\mathrm{Rings} \cong \mathbb{Z}\text{-alg}$ fully faithfully embeds in the category of $H\mathbb{Z}$ -algebras we can see that two rings are isomorphic iff their corresponding $H\mathbb{Z}$ algebras are isomorphic as $H\mathbb{Z}$ algebras.

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